Rahul Vishnuvardhana

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Education

Northeastern University, Boston, MA
Master of Science in Electrical and Computer Engineering (GPA: 3.8) *Expected May 2028*
Concentration: Hardware and Software in Machine Intelligence
Coursework: Machine Learning; Large Language Models; Reinforcement Learning; Fundamentals of Computer Engineering

Sri Sairam Engineering College, Anna University, India
Bachelor of Engineering in Electronics and Communication Engineering *April 2022*
Publication: Machine learning based image analysis for grain quality assessment, IJETMS Vol. 6, July 2022

Technical Skills

Programming: Python; SQL; TypeScript; Bash; data structures; algorithms
Machine Learning: PyTorch; LSTM; multi-head attention; focal loss; conformal prediction; ablation studies
Artificial Intelligence: LLMs; RAG; agentic frameworks; prompt engineering; vector search
Data and Orchestration: Snowflake; Oracle; PostgreSQL; ETL; database migrations; AutoSys; Redwood
Development: REST APIs; FastAPI; Next.js; Docker; Git; GitHub Copilot; Linux; Tableau
Cloud Certificates: Azure Data Engineer Associate (DP-203); Azure Fundamentals (AZ-900)

Professional Experience

Hexaware Technologies Private Limited, India (Client: Brown Brothers Harriman)
Software Engineer, Data Warehouse and Reporting team *September 2022 to September 2025*

- Improved average incident resolution time by 50% across 50+ business-critical AutoSys and Redwood pipelines on Linux, instrumenting each job with structured logging and tracing failures through execution metrics to root cause.
- Held business-reported data discrepancies to a maximum of 5 per month with Python services that validated API payloads and CSV feeds at ingest and reconciled them against source using row counts and checksums.
- Designed ETL pipelines and database migrations in Python, SQL and Oracle Data Integrator that consolidated 8+ source systems into analytics-ready tables, turning an hour-long manual report into automated same-day delivery.
- Refactored one-off scripts into reusable object-oriented components with help from GitHub Copilot and LLM assistants, standardizing configuration and database connections across the codebase.
- Eliminated a single-person dependency across 250+ processes through runbooks, standard operating procedures and Tableau dashboards that let teammates and stakeholders self-serve.

Graduate Engineer Trainee *March 2022 to August 2022*

- Applied Python, SQL, Snowflake and Oracle across the graduate program, advancing into a specialized Data Engineer track within 3 months, ahead of schedule.

Projects

LOEWISE: Independent Project, In Progress *Fall 2026*
Retrieval system that answers questions across FDA drug and patent records

- Prototyping an agentic retrieval pipeline in Python over FDA Orange Book and patent records, answering drug patent questions from cited sources instead of scattered documents.
- Testing each answer against its source document and flagging low-confidence extractions for human review, keeping unverified answers away from users.
- Cross-referencing Orange Book patent use codes with FDA labeling to flag where coverage exceeds approved uses.

TraceOp: Independent Project *Summer 2026*
Copilot and outreach tracker for research and job roles | v0-traceop.vercel.app

- Integrated live postings from the Greenhouse, Lever, Ashby and Workday boards plus OpenAlex, NSF and NIH into a Next.js, TypeScript and Supabase application, surfacing **2,693 faculty across 7 colleges** in one search.
- Implemented pgvector retrieval so Gemini drafts never invent a skill, metric or date, with Vitest and Playwright tests.

LEO Gateway: Independent Project *Spring 2026*
Failover gateway that keeps LLM API traffic alive through provider outages | leo-api-intelligence.onrender.com

- Developed a Python reverse-proxy gateway on FastAPI and Docker that detects 5xx, 429 and timeout errors and retries in-flight requests on a healthy provider, so applications keep working through outages.
- Trained a bidirectional LSTM in PyTorch with multi-head attention and focal loss on 1.94M rows to **ROC-AUC 0.8202** and PR-AUC 0.5383, predicting failures early enough to reroute traffic.
- Calibrated uncertainty with conformal prediction and ran ablation studies to isolate what drove performance, making each risk score reliable enough to deploy live.